THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The polar regions are colder than the equator. Reason (R): Sunlight falls at a slant at the poles and is spread over a larger area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A surface of albedo 0.5 receives 800 W/m squared. The absorbed radiation per square metre in 10 s is: 1

(a) 8000 J **(b)** 400 J **(c)** 800 J **(d)** 4000 J

3. A farmer applies too much nitrogen fertiliser. The most likely consequence downstream is: 1

(a) clear water **(b)** algal bloom and fish kill
(c) more fish **(d)** more oxygen

4. The troposphere is warmer near the ground mainly because: 1

(a) the ground is heated by sunlight and warms the air
(b) the Sun is closer
(c) ozone absorbs UV
(d) greenhouse gases are absent

5. Assertion (A): Melting of polar ice accelerates global warming. Reason (R): Water has a lower albedo than ice. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. If the albedo of the Earth decreases, the Earth's temperature will: 1

(a) decrease **(b)** increase **(c)** remain the same **(d)** become zero

7. A wetland is drained for building. Which service of the Earth system is lost? 1

(a) photosynthesis **(b)** water filtration and flood control
(c) ozone **(d)** gravity

8. The stratosphere is warmer at the top due to the: 1

(a) ground heating **(b)** absorption of UV by ozone
(c) volcanoes **(d)** oceans

9. Assertion (A): The natural greenhouse effect is essential for life on Earth. Reason (R): Without it the Earth would be too cold to support life. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A tree absorbs 22 kg of CO_2 per year. The number of trees needed to offset the 2200 kg emission of a family is: 1

(a) 100 **(b)** 10 **(c)** 22 **(d)** 1000

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The average albedo of the Earth is 0.3. Calculate the radiation absorbed per square metre if the average incident is 342 W/m squared. 2

12. State why India is vulnerable to climate change. Give two reasons. 2

13. Why does the stratospheric ozone absorb UV but not visible light? 2

14. State two feedback loops that can accelerate global warming. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A household uses 250 units per month. Using 0.8 kg CO_2 per unit, calculate the annual emission and the number of trees (each absorbing 22 kg per year) needed to offset it. 3

16. Why do deserts have large day-night temperature differences? Use the ideas of albedo and water vapour. 3

17. Describe how energy flows in the Earth system, from the Sun to the heat radiated to space, taking 100 units reaching the Earth with 30 units reflected. 3

OR

Calculate the energy and CO_2 saved if a school replaces 100 tube lights of 40 W by 20 W LED tubes used 8 hours a day for 250 days (0.8 kg per kWh). 3

18. Explain the importance of rainforests as 'lungs of the Earth' and the problems of treating them as such with a better description. 3

OR

Explain why using fossil fuels disturbs the carbon cycle's balance. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A student models the Earth's temperature by the balance of incoming and outgoing energy. The Sun delivers 342 W/m squared on average, the albedo is 0.3. Calculate the absorbed energy, the outgoing energy needed at equilibrium and state what happens if the albedo decreases to 0.28 and the effect of more greenhouse gases. 5

OR

Explain how deforestation affects the water cycle, the carbon cycle, the albedo and the biodiversity in a region with examples from the Amazon or the Western Ghats. 5

20. Explain how the production of 1 tonne of cement or steel releases CO_2 and how the use of renewable energy and recycling can reduce the emissions, with numbers (cement about 0.9 tonne CO_2 per tonne). 5

OR

A town gets 800 mm of rain per year over an area of 20 square km. Calculate the volume of rainfall in cubic metres, the amount that can be harvested with 60 per cent efficiency and discuss how it solves the water shortage if the town uses 5 million cubic metres a year. 5

END OF QUESTION PAPER Total: 40 Marks